Subject Description Form

Subject Code	COMP5424
Subject Title	Extended Reality
Credit Value	3
Level	5
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	The objectives of this subject are to: 1. provide students with an in-depth view of extended reality (XR) and relevant interactive technologies including selected topics of the latest trends in both academic and industrial contexts; 2. equip students with interdisciplinary knowledge regarding both the technological and human aspects of XR; 3. provide students with the knowhow of user experience design for XR applications; 4. equip students with knowledge and skills in design, development, and evaluation of XR applications; and 5. nurture students' humanistic thinking and aesthetic sense in the design of XR applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: *Professional/academic knowledge and skills* a) develop a comprehensive understanding of the key theories and concepts in XR; b) demonstrate advanced knowledge and practical skills in XR and related interactive technologies; c) apply advanced knowledge and practical skills creatively in the design of XR applications to address real-world challenges; d) critically evaluate XR applications from the perspectives of academic research and industrial practice through empirical approaches; e) critically synthesize design ideas from the latest trends and emerging challenges in XR; and f) incorporate humanistic perspectives and aesthetic considerations in the design of XR and related interactive applications.

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<td><u>Attributes for all-roundedness</u>
g) effectively collaborate in teams by employing systematic approaches and appropriate project management strategies;
h) rapidly develop prototypes for XR and related interactive applications; and
i) effectively communicate design concepts and development processes to diverse audiences.</td>
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<td>Subject Synopsis/ Indicative Syllabus</td>
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<tr><td>Topic</td></tr>
<tr><td>1. Introduction to XR
Definitions of XR; historical context of XR (analogue & digital); interdisciplinary nature of XR.</td></tr>
<tr><td>2. Perceptual, Cognitive and Psychological Aspects of XR
Depth perception; color perception; auditory and vestibular systems; sensorimotor contingency; motion sickness and simulator sickness; definition of immersion; hardware design; presence and immersion; place illusion; plausibility illusion; rubber hand illusion and the sense of embodiment; self-avatar.</td></tr>
<tr><td>3. Spatial User Interface and User Experience Design for XR
Fitts' law and Fitts' law in spatial user interface; common user interface design decisions in XR applications; basics of user experience design; accessibility of spatial user interfaces; hybrid user interface.</td></tr>
<tr><td>4. Advanced Visual Rendering and Multisensory XR
Rasterization; raytracing; photogrammetry; neural rendering; foveated rendering; spatial audio; haptic rendering.</td></tr>
<tr><td>5. XR Application Design and Development
Exemplary applications; development process and cycles; co-design; participatory design; low-resolution and high-resolution prototyping.</td></tr>
<tr><td>6. Application Evaluations
Technology acceptance; importance of evaluation; user study and experiment design; quantitative and qualitative evaluation techniques.</td></tr>
<tr><td>7. Latest Trends and Challenges in XR
Multimodal Interfaces and Tangible Interfaces
Spatial audio and psychoacoustics; rendering techniques; raytracing; haptics and tactile; pseudo haptics; sensor fusion; tangible interface and haptic retargeting; exemplary applications;
Collaborative Virtual Environment and social XR
Co-presence, tele-presence, and social presence; the uncanny valley hypothesis; embodied agents and self-avatars; collaborative interactions; social XR and Metaverse; affective computing; privacy and ethical issues.</td></tr>
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Teaching/ Learning Methodology

Lectures, Tutorials and Labs

The subject material will be delivered through lectures, tutorials, and labs. Lectures will focus on the delivery of the theories, definitions, facts, and guidelines with examples from both academic and industrial contexts. Guest lectures from academic and/or industry will be invited to introduce the latest trends in XR research and practice. Tutorials and labs will provide students with knowhow in de-facto standards and platforms for XR application development. Tutorials and labs, employing the blended learning approach of delivery, will also help students gain hands-on experiences in design, development, and evaluation of XR applications in groups.

Group Project

The group project will provide students with in-depth opportunities to practice the lecture concepts, as well as to assess their ability to apply these concepts in practical scenarios of creating XR applications collaboratively.

Examination

The final examination will assess students on their grasp of the subject materials.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed								
		a	b	c	d	e	f	g	h	i
Continuous Assessment	**60%**	✓	✓	✓	✓	✓	✓	✓	✓	✓
1. Mid-term Test	20%	✓	✓	✓	✓	✓	✓			
2. Group project	40%	✓	✓	✓	✓	✓	✓	✓	✓	✓
Final Examination	**40%**	✓	✓	✓	✓	✓	✓			
Total	100%									

The course will be accessed by group project, mid-term test, and the final examination.

The mid-term test is employed to assess students' ability in understanding key concepts of XR and applying these key concepts to address XR application design challenges. The Group project is used to develop students' ability in solving problems by using systematic approaches, collaboration with peer students, and quick prototyping of XR applications using de-facto standards and platforms when facing real-world scenarios. Individual contributions to the group project will be evaluated through self-reported contribution lists and workload distribution lists. The final examination is used to assess students on their grasp of the subject materials.

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<tr><td rowspan="6">Student Study Effort Expected</td><td colspan="2">Class contact:</td></tr>
<tr><td>▪ Lectures, Tutorials, and Labs</td><td>39 Hrs.</td></tr>
<tr><td colspan="2">Other student study effort:</td></tr>
<tr><td>▪ Group Project, Self-study for Mid-term Test and Final Examination</td><td>76 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>115 Hrs.</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Reading List and References</td><td colspan="2">Textbook:
1. LaValle, S. M. (2016). Virtual Reality. Cambridge University Press.
2. LaViola, J., Kruijff, E., Bowman, D., McMahan, R., & Poupyrev, I. (2017). 3D User Interfaces: Theory and Practice. Addison-Wesley Professional.

Reference Books:
3. Gregory, J. (2018). Game Engine Architecture, Third Edition. AK Peters/CRC Press..
4. Aron, A., & Aron, E. N. (2012). Statistics for Psychology 6th Edition. Pearson.
5. Khronos Group. (2025). OpenXR API Registry. Retrieved from https://registry.khronos.org/OpenXR/

Reading List:
6. Al Zayer, M., MacNeilage, P., & Folmer, E. (2018). Virtual locomotion: a survey. IEEE transactions on visualization and computer graphics, 26(6), 2315-2334. https://doi.org/10.1109/TVCG.2018.2887379
7. Clark, L. D., Bhagat, A. B., & Riggs, S. L. (2020). Extending Fitts' law in three-dimensional virtual environments with current low-cost virtual reality technology. International journal of human-computer studies, 139, 102413. https://doi.org/10.1016/j.ijhcs.2020.102413
8. Kilteni, K., Groten, R., & Slater, M. (2012). The sense of embodiment in virtual reality. Presence: Teleoperators and Virtual Environments, 21(4), 373-387. https://doi.org/10.1162/PRES_a_00124
9. Obrenovic, Z., Abascal, J., & Starcevic, D. (2007). Universal accessibility as a multimodal design issue. Communications of the ACM, 50(5), 83-88.
10. Skarbez, R., Brooks, Jr, F. P., & Whitton, M. C. (2017). A survey of presence and related concepts. ACM Computing Surveys (CSUR), 50(6), 1-39. https://doi.org/10.1145/3134301</td></tr>
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